Assessing Studies Based on Multiple Regression

SW Chapter 9

ECON3500: Econometrics and Applications

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Learning Objectives

By the end of this chapter, you will be able to:

Define **internal validity** and **external validity**

Identify five threats to internal validity

Explain how each threat causes bias or incorrect inference

Propose solutions to each threat

Distinguish between **causal inference** and **forecasting** objectives

Internal and External Validity

What Is Validity?

Validity = Do we believe the results of our estimation?

Did we learn what we set out to learn?

Did we estimate the causal impact of class size on test scores?

Did we correctly measure the association between marital status and wages?

We need a **framework** for evaluating whether a study is useful for answering a particular question.

What are the **threats to validity**?

Two Dimensions of Validity

Internal Validity

The statistical inferences about causal effects are valid **for the population being studied**.

External Validity

The statistical inferences can be **generalized** from the population and setting studied to other populations and settings.

“Setting” = the legal, policy, and physical environment and related salient features.

Threats to External Validity

Assessing external validity requires **detailed substantive knowledge** and judgment on a case-by-case basis.

Example: How far can we generalize class size results from California?

Differences in populations

- California in 2011? Massachusetts in 2011? Mexico in 2011?

Differences in settings

- Different legal requirements (e.g., special education)
- Different treatment of bilingual education

Differences in inputs

- Teacher characteristics, school resources, curricula

There is no mechanical test for external validity — it requires **judgment**.

Internal Validity: Two Requirements

Estimates are internally valid if:

The estimator is **consistent** (or unbiased)

Confidence intervals have **correct coverage** — in at least 95% of samples, the 95% CI contains the true parameter

The Rest of This Chapter

We focus on threats to **internal validity**: things that can make our regression estimates misleading for the population we're studying.

Five Threats to Internal Validity

Overview

Five threats to the internal validity of regression studies:

Omitted variable bias

Wrong functional form

Errors-in-variables bias (measurement error)

Sample selection bias

Simultaneous causality bias

All five imply that $E(u_i | X_{1i}, \ldots, X_{ki}) \neq 0$

$\Rightarrow$ Conditional mean independence fails

$\Rightarrow$ OLS is **biased and inconsistent**

Three Categories of Threats

When you encounter a potential threat, classify it:

Category	What goes wrong	Severity
1. Affects **consistency** of $\hat{\beta}$	Estimator converges to wrong value	Most serious
2. Affects **inference** but not consistency	CIs have wrong coverage; $\hat{\beta}$ is still consistent	Serious
3. Increases **imprecision** only	Larger SEs, but $\hat{\beta}$ unbiased and CIs correct	Least serious

Categories 1 and 2 are **actual threats** to validity

Category 3 makes estimates imprecise but standard errors properly reflect that imprecision

Knowledge Check

A researcher drops half the sample at random due to a data processing error. Which category is this?

. . .

Category 3 — smaller sample means larger SEs, but no bias. (This is like Case 1 of missing data, which we'll see shortly.)

Threat 1: Omitted Variable Bias

OVB: We Know This One!

Omitted variable bias arises if an omitted variable is both:

A **determinant of Y**, and

Correlated with at least one included regressor

The question to ask:

With control variables included, are there **still** omitted factors that are:

Not adequately controlled for?

Correlated with the variable of interest?

Solutions to OVB

If the omitted variable **can be measured**: include it as an additional regressor

If you have **adequate controls** (conditional mean independence plausibly holds): include them

Run a **randomized controlled experiment**

- If X is randomly assigned, then X is distributed independently of u, so $E(u|X) = 0$

Coming later in this course:

Use **panel data** where each entity is observed more than once

Use **instrumental variables** regression

Threat 2: Wrong Functional Form

Functional Form Misspecification

Arises if the functional form is incorrect — for example, an interaction term or quadratic is incorrectly omitted.

Solutions:

Continuous Y: Use appropriate nonlinear specifications (logarithms, quadratics, interactions — Ch8)

Discrete Y (e.g., binary): Use probit or logit (beyond ECON3500)

> **Perspective**
>
> Functional form misspecification is rarely the **biggest** problem in regression analysis. It matters, but OVB and measurement error are usually more concerning.

Threat 3: Errors-in-Variables Bias

Measurement Error in Economic Data

So far we've assumed X is measured without error. In reality, economic data often have measurement error:

Data entry errors in administrative data

Recollection errors in surveys ("When did you start your current job?")

Ambiguous questions ("What was your income last year?" — before or after tax?)

Intentionally false responses ("What is the current value of your financial assets?" "How often do you drink and drive?")

Where and What Kind?

Measurement error can occur in the **dependent variable** or **independent variables**.

And it can be **classical** (random) or **non-classical** (systematic).

	Classical (random)	Non-classical (systematic)
Dependent variable	Minor problem — larger variance	Serious problem — bias
Independent variable	**Attenuation bias**	Serious problem — bias

Classical Errors-in-Variables (CEV) Assumption

The measurement error is uncorrelated with the true value: $\text{Cov}(w_i, X_i^*) = 0$

Measurement Error in Y

Suppose we observe y_i but the true value is y_i^*:

$$y_i = y_i^* + e_0$$

The population regression is:

$$y_i^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u_i$$

What we actually estimate:

$$y_i = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \underbrace{(u_i + e_0)}_{\text{new error term}}$$

If e_0 is uncorrelated with the X's, the **coefficients are still unbiased** — the error term just has more variance, so standard errors are larger.

Measurement error in Y under CEV: not a big deal.

Measurement Error in X: Setup

The true model is:

$$Y_i = \beta_0 + \beta_1 X_i^* + u_i$$

We observe $X_i = X_i^* + w_i$ instead of X_i^*, where $E[w_i] = 0$.

Assumptions (classical errors-in-variables):

$\text{Cov}(w_i, u_i) = 0$ — measurement error uncorrelated with other omitted factors

$\text{Cov}(w_i, X_i^*) = 0$ — measurement error uncorrelated with the true value $\leftarrow$ **the CEV assumption**

$\text{Cov}(X_i^*, u_i) = 0$ — no traditional OVB

Why Is This a Problem?

The key insight: even though $\text{Cov}(w_i, X_i^*) = 0$, the measurement error w_i **is** correlated with the observed X_i.

$$\text{Cov}(X_i, w_i) = \text{Cov}(X_i^* + w_i,\ w_i)$$

$$= \underbrace{\text{Cov}(X_i^*, w_i)}_{=\ 0 \text{ by CEV}} + \text{Cov}(w_i, w_i)$$

$$= \text{Var}(w_i) = \sigma_w^2$$

The observed regressor X_i is correlated with the measurement error, which is part of the composite error term. This **violates conditional mean independence**.

Derivation of Attenuation Bias

$$\hat{\beta_1} \xrightarrow{p} \frac{\mathrm{Cov}(X_i, Y_i)}{\mathrm{Var}(X_i)}$$

$$= \frac{\mathrm{Cov}(X_i^* + w_i,\ \beta_0 + \beta_1 X^* + u_i)}{\mathrm{Var}(X_i^* + w_i)}$$

Expanding the numerator:

$$= \frac{\mathrm{Cov}(X_i^*, \beta_0) + \mathrm{Cov}(X_i^*, \beta_1 X_i^*) + \mathrm{Cov}(w_i, \beta_0) + \mathrm{Cov}(w_i, \beta_1 X^*) + \mathrm{Cov}(w_i, u_i)}{\mathrm{Var}(X_i^*) + \mathrm{Var}(w_i) + 2\mathrm{Cov}(X_i^*, w_i)}$$

Applying our assumptions ($\mathrm{Cov}(w, X^*) = 0$, $\mathrm{Cov}(w, u) = 0$):

$$= \frac{0 + \beta_1 \mathrm{Var}(X_i^*) + 0 + 0 + 0}{\mathrm{Var}(X_i^*) + \mathrm{Var}(w_i) + 0}$$

$$= \beta_1 \cdot \frac{\mathrm{Var}(X_i^*)}{\mathrm{Var}(X_i^*) + \mathrm{Var}(w_i)}$$

Attenuation Bias: The Result

$$\hat{\beta_1} \xrightarrow{p} \beta_1 \cdot \frac{\text{Var}(X_i^*)}{\text{Var}(X_i^*) + \text{Var}(w_i)}$$

The fraction on the right is **between 0 and 1**. So:

If $\beta_1 > 0$: estimate is biased **toward zero** (too small)

If $\beta_1 < 0$: estimate is biased **toward zero** (too close to zero)

Attenuation Bias

Classical measurement error in a regressor always biases the coefficient **toward zero** — making the estimated effect look **smaller** than it truly is.

Attenuation Bias: Intuition

Why does this happen?

X_i has a **higher variance** than the true X_i^* (the noise adds spread)

When X_i moves, Y doesn't always respond — because Y only responds to movements in X^*

The regression sees changes in X that are **not related to** Y

It concludes: "the effect of X on Y must be small"

Practical implication: If we think there's measurement error in X, our estimate is a **lower bound** on the true magnitude.

What If CEV Doesn't Hold?

Sometimes the classical assumption is unlikely:

People with high incomes **under-report** and people with low incomes over-report

- $\Rightarrow \text{Cov}(\text{income}, \text{error}) \neq 0$

People are less accurate about their age as they get older

- $\Rightarrow \text{Cov}(\text{age}, \text{error}) \neq 0$

When CEV fails:

The direction of bias is **ambiguous** — no longer necessarily toward zero

Interpreting the impact is more complicated (beyond ECON3500)

Living with Measurement Error

Practical Advice

Dependent variable (classical): Don't worry too much — estimates still unbiased

Independent variable (classical): We can still sign our magnitudes

- e.g., "Returns to education are **at least** 10%"

Non-classical: More serious — consider better data or instrumental variables

Better data always helps!

Use administrative records instead of survey self-reports when possible.

Threat 4: Sample Selection Bias

Missing Data: Three Cases

Data are often missing. Whether this introduces bias depends on **why** the data are missing:

Case	Data missing because of…	Bias?
1	**Random chance**	No
2	**Values of X**	No
3	**Values of Y or u**	**Yes**

Cases 1 and 2: standard errors are larger, but $\hat{\beta}$ is **unbiased**

Case 3: introduces **sample selection bias**

Case 1: Missing at Random

Suppose you survey 100 workers and record answers on paper — but your dog eats 20 response sheets (selected at random) before you enter them.

This is equivalent to having a random sample of 80 workers.

Your dog didn't introduce any bias!

Knowledge Check

A server crash deletes 30% of your dataset at random. Should you worry about bias?

. . .

No — this is Case 1. You lose precision (larger SEs) but no bias.

Case 2: Missing Based on X

In the test score application, suppose you restrict analysis to districts with $STR < 20$.

You can't say anything about districts with large class sizes

But focusing on small-class districts **doesn't introduce bias**

This is equivalent to having missing data where data are missing if $STR > 20$

More generally: if data are missing based **only on values of** X, the missing data don't bias OLS.

Case 3: Missing Based on Y or u

This **does** introduce bias. This is **sample selection bias**.

Sample selection bias arises when a selection process:

Influences the **availability of data**, and

Is **related to the dependent variable**

Example: Height of Undergraduates

You want to estimate the mean height of undergraduates.

You collect data by standing outside the **basketball team's locker room** and recording the height of undergraduates who enter.

Is this a good design?

You've sampled individuals in a way that is **related to the outcome** Y (height)

This results in bias — your estimate will be too high

Example: Mutual Fund Performance

Question: Do actively managed mutual funds outperform "hold-the-market" index funds?

Design:

Sample: mutual funds available to the public **today**

Data: returns for the preceding 10 years

Compare: average 10-year return vs. S&P 500

Is there sample selection bias?

Yes — funds that performed poorly **went out of business** and are not in today's sample.

The surviving funds are the "basketball players" of mutual funds.

This is called **survivorship bias**.

Why Sample Selection Biases OLS

Sample selection creates a direct correlation between a regressor and the error term.

Consider the mutual fund regression:

$$\text{return}_i = \beta_0 + \beta_1\ \text{managedfund}_i + u_i$$

$\text{managedfund}_i = 1$ for actively managed funds in our sample (i.e., the survivors).

Being managed *and in our sample* means the fund outperformed failed managed funds — which are not in the sample.

$$\Rightarrow \text{Cov}(\text{managedfund}_i,\ u_i) \neq 0$$

The surviving managed funds are the "basketball players" of mutual funds — selected **because of** their returns.

⇒ Conditional mean independence fails ⇒ **OLS is biased**

Example: Returns to Education

Question: What is the return to an additional year of education among college graduates?

Design:

Sample: **employed** college graduates (so we have wage data)

Regress: $\ln(\text{earnings})$ on years of education

Is there sample selection bias?

Yes — we only observe wages for people who are **employed**. Employment is related to the outcome (wages/earnings potential), so the sample is not representative.

Solutions to Sample Selection Bias

Collect data properly — design sampling to avoid selection

- *Basketball:* true random sample from enrollment records
- *Mutual funds:* sample funds available at the **beginning** of the period (include failed funds)
- *Returns to education:* sample college graduates, not just employed workers

Randomized controlled experiment

Model the selection process and estimate it (advanced — not in ECON3500)

Threat 5: Simultaneous Causality Bias

When Y Causes X Too

So far we've assumed X causes Y. But what if Y **also causes** X?

This is **simultaneous causality** (or "reverse causality").

Example: Class size effect

Low STR $\longrightarrow$ better test scores (the effect we want)

But: districts with **low test scores** may receive extra resources $\longrightarrow$ also get low STR (political process)

What does this mean for a regression of TestScore on STR?

The coefficient on STR reflects **both** directions of causality — it's biased.

Solutions to Simultaneous Causality Bias

Randomized controlled experiment — because X_i is chosen at random, there is no feedback from Y to X (assuming perfect compliance)

Develop a complete model of both directions of causality — this is the approach behind large macroeconomic models (e.g., Federal Reserve). **Extremely difficult in practice.**

Instrumental variables regression — estimate the causal effect of X on Y while ignoring the feedback from Y to X (coming later!)

Putting It All Together

Summary: Five Threats to Internal Validity

Threat	What Happens	Consequence	Solutions
1. **OVB**	Omitted variable correlated with X and determines Y	Biased, inconsistent	Add controls, RCT, IV, panel data
2. **Wrong functional form**	Incorrect specification (missing interactions, nonlinearities)	Biased	Logs, quadratics, interactions (Ch8)
3. **Measurement error**	X or Y measured with error	Attenuation bias (classical); ambiguous bias (non-classical)	Better data, IV
4. **Sample selection**	Data missing based on Y or u	Biased, inconsistent	Better sampling design, RCT
5. **Simultaneous causality**	Y causes X and X causes Y	Biased, inconsistent	RCT, IV

Common Thread

All five threats cause $E(u_i|X_{1i},\ldots,X_{ki}) \neq 0$ — conditional mean independence fails, and OLS is biased.

Decision Tree: Is Missing Data a Problem?

Step 1: Why are the data missing?

Random chance (server crash, lost surveys, random attrition)

- $\longrightarrow$ **No bias.** Smaller sample $\longrightarrow$ larger SEs. Proceed.

Based on X values (restricted sample, subgroup analysis)

- $\longrightarrow$ **No bias** for the subpopulation. Can't generalize beyond it (external validity concern).

Based on Y or u (survivorship, employment requirement, self-selection)

- $\longrightarrow$ **Sample selection bias!** Redesign sampling or use specialized methods.

The Key Question

Ask yourself: "Would the **reason** someone is missing from my data be correlated with the **outcome** I'm trying to measure?"

If yes $\longrightarrow$ sample selection bias.

Forecasting vs. Causal Inference

Two Very Different Objectives

	Causal Inference	Forecasting
Goal	Estimate effect of changing X on Y	Predict Y given observed X's
What matters most	Unbiased $\hat{\beta}$	Good fit ($\bar{R}^2$)
OVB	**Critical problem**	Not a problem!
Coefficient interpretation	Very important	Not important
External validity	Important for generalization	**Paramount** — model must hold in new data

Forecasting: A Different Mindset

When the goal is **forecasting**:

$\bar{R}^2$ matters — a lot!

Omitted variable bias **isn't a problem** — we don't care about the causal interpretation of individual coefficients

The important thing is a **good fit** and a model you can "trust" in your application

But **external validity is paramount**: the model estimated using historical data must hold into the (near) future.

> ⚠ **Don't Confuse the Two**
>
> A model with high R^2 and biased coefficients might be great for forecasting but useless for policy. A model with unbiased coefficients and low R^2 might be ideal for causal inference but poor for prediction.

Knowledge Check: Identify the Threat

For each scenario, identify the primary threat to internal validity:

1. A study of returns to education uses IQ as a proxy for ability, but IQ tests are noisy measures of true ability.

$\longrightarrow$ **Errors-in-variables bias** (measurement error in X) — attenuation bias on the IQ coefficient

2. A study of the effect of police on crime finds that cities with more police have more crime.

$\longrightarrow$ **Simultaneous causality** — high-crime cities hire more police

3. A study of job training effects on wages only observes wages for people who are employed after training.

$\longrightarrow$ **Sample selection bias** — employment status is related to the outcome (wages)

Key Takeaways

Internal validity = inferences are correct for the population studied; **external validity** = results generalize to other settings

Five threats all cause the same fundamental problem: $E(u|X) \neq 0$

Measurement error in X (classical) causes attenuation bias — estimates are too small in magnitude

Sample selection bias arises when data are missing because of Y, not because of X or random chance

Simultaneous causality is hard to solve without experiments or instrumental variables

Forecasting and causal inference have different validity concerns — don't confuse them

Looking Ahead

Many of these threats will motivate the methods we learn next: **panel data** (Ch10) and **instrumental variables** (Ch12) are direct responses to OVB and simultaneous causality.